

# FROZEN CONFIGURATIONS OF A SHIFTED FREDKIN BLOCK NETWORK: A TRANSFER MATRIX, CLOSED COUNTS, AND A SPATIAL ZETA FUNCTION

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ABSTRACT. We study a binary ring of length  $3m$  updated by two staggered layers of control-on-one three-bit Fredkin gates. The first layer acts on aligned triples; the second acts on triples shifted by one site. Although the full map is a Hamming-weight-preserving permutation, its frozen configurations admit a rank-two transfer description. We prove that they form a nearest-neighbor shift of finite type on the eight three-bit blocks, with an explicit adjacency matrix  $M$  satisfying

$$\det(\lambda I - M) = \lambda^6(\lambda^2 - 5\lambda + 3).$$

Consequently, the number  $f_m$  of frozen configurations is

$$f_m = \left(\frac{5 + \sqrt{13}}{2}\right)^m + \left(\frac{5 - \sqrt{13}}{2}\right)^m,$$

and obeys  $f_m = 5f_{m-1} - 3f_{m-2}$  with  $f_1 = 5$  and  $f_2 = 19$ . The associated *spatial* frozen-set zeta function is  $(1 - 5z + 3z^2)^{-1}$ . This is the zeta function of the block shift on the frozen subshift, not a temporal zeta function of the finite-ring update. Exhaustive evaluation of all 299,592 states for  $m \leq 6$  checks the transfer description, reversibility, conservation, and a maximum observed temporal period of 74; no all-size temporal-period claim is made.

## 1. INTRODUCTION

The Fredkin controlled-swap gate is a basic conservative reversible-logic element: it permutes two target bits conditional on a control bit and preserves the total number of ones. The gate and its role in conservative computation belong to Fredkin and Toffoli's foundational work [1]. Alternating block partitions are likewise a standard route to globally reversible cellular dynamics [2, 6]. These two ingredients make finite permutations easy to construct, but they do not by themselves describe the permutations' frozen configurations.

This note considers one concrete one-dimensional construction: apply the control-on-one three-bit Fredkin gate first on aligned triples and then on triples shifted by one site. Its fixed-point equations are local only after the ring is grouped into three-bit blocks. In that representation they become an eight-symbol nearest-neighbor constraint. The resulting adjacency matrix has rank two, which turns the frozen-state count into a second-order recurrence.

The exact contributions are as follows.

- (1) We record the global inverse, an explicit reversing involution, and Hamming-weight conservation for every ring size  $3m$ .
- (2) We prove an if-and-only-if local characterization of frozen configurations and give its  $8 \times 8$  transfer matrix.
- (3) We derive the closed fixed-state count, recurrence, frozen-set entropy, and the spatial zeta function  $(1 - 5z + 3z^2)^{-1}$ .
- (4) A literal-rule exhaustive control checks every state through  $m = 6$  and reports the finite temporal cycle census without extrapolating it to all  $m$ .

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The distinction between space and time is essential. For each  $m$ , the finite-ring update is a permutation  $T_m$  on  $2^{3m}$  states. Our zeta function instead counts configurations periodic under translation by one *three-site block* inside the infinite frozen set. We neither compute the Artin–Mazur zeta function of temporal iteration of  $T_m$  nor claim that the observed temporal periods are unbounded.

## 2. PRIOR OWNERSHIP AND RESIDUAL SCOPE

Fredkin and Toffoli introduced conservative logic and the controlled interchange primitive [1]. Toffoli and Margolus developed block cellular-automaton architectures in which bijective local maps produce reversible global updates [6]; Kari later gave structural results for reversible cellular automata represented by overlapping block partitions [2]. Morita surveys the realization of Fredkin gates in several simple reversible cellular automata [4]. We treat all of this architecture-level material as prior ownership.

The closest dynamical comparison is the Fredkin staircase automaton of Singh, Vasseur, and Gopalakrishnan [5]. Their Floquet circuit uses three layers of four-site constrained-swap gates and is analyzed through quasiparticles, transport, and integrability. The present update uses two layers of the control-on-one three-bit controlled-swap gate and asks a static enumeration question about its frozen set. Thus the residual object here is the transfer matrix and exact frozen-state package for the specific map in [definition 3.2](#), not Fredkin dynamics, conservative logic, reversible block cellular automata, or integrability in general.

No absolute priority claim is made. In particular, this note does not claim the first Fredkin-based cellular automaton, the first reversible conservative network, or the first staggered controlled-swap circuit. The literature boundary above is part of the theorem contract rather than an after-the-fact qualification.

## 3. THE SHIFTED TWO-LAYER MAP

Write  $\mathbb{Z}_r = \mathbb{Z}/r\mathbb{Z}$ . A state on the ring of length  $3m$  is  $x = (x_i)_{i \in \mathbb{Z}_{3m}} \in \{0, 1\}^{\mathbb{Z}_{3m}}$ . All site indices below are taken modulo  $3m$ .

**Definition 3.1** (control-on-one Fredkin gate). For  $(c, u, v) \in \{0, 1\}^3$ , set

$$F(c, u, v) = \begin{cases} (c, u, v), & c = 0, \\ (c, v, u), & c = 1. \end{cases}$$

The first coordinate is the control and the last two are the targets. Fredkin and Toffoli’s original table uses the complementary control-on-zero convention [1]; control-on-one is now also common. The explicit definition above fixes the convention used throughout.

**Definition 3.2** (two-layer shifted Fredkin map). Fix  $m \geq 1$ . Let  $A_m$  apply  $F$  in parallel to the disjoint triples

$$(3j, 3j + 1, 3j + 2), \quad j \in \mathbb{Z}_m,$$

and let  $B_m$  apply  $F$  in parallel to

$$(3j + 1, 3j + 2, 3j + 3), \quad j \in \mathbb{Z}_m.$$

One full time step is

$$T_m = B_m \circ A_m.$$

The frozen set is  $\mathcal{F}_m = \text{Fix}(T_m)$ .

Each family of triples in a single layer partitions the ring, so the parallel updates are unambiguous. The definition includes  $m = 1$ : the second layer then sees the same three sites in cyclically shifted roles.

**Proposition 3.3** (reversibility and conservation). *For every  $m \geq 1$ , the maps  $A_m$  and  $B_m$  are involutions. Consequently,  $T_m$  is a permutation with*

$$T_m^{-1} = A_m \circ B_m, \quad A_m \circ T_m \circ A_m = T_m^{-1}.$$

Moreover,  $T_m$  preserves the Hamming weight  $|x|_1 = \sum_{i \in \mathbb{Z}_{3m}} x_i$ .

*Proof.* The gate  $F$  is an involution because applying the same conditional swap twice restores the two targets. Gates within one layer have disjoint supports, so the product defining each layer is also an involution. Hence  $(B_m A_m)^{-1} = A_m B_m$ , and

$$A_m(B_m A_m)A_m = A_m B_m = (B_m A_m)^{-1}.$$

Every local gate only permutes its two target coordinates and leaves its control unchanged. It therefore preserves the number of ones on its support. Both layers, and thus their composition, preserve the global Hamming weight.  $\square$

#### 4. FROZEN CONFIGURATIONS AS A NEAREST-NEIGHBOR SFT

Group the sites into blocks

$$s_j = (a_j, b_j, c_j) = (x_{3j}, x_{3j+1}, x_{3j+2}) \in \{0, 1\}^3$$

and index a block by

$$\iota(a, b, c) = a + 2b + 4c \in \{0, \dots, 7\}.$$

Thus index order is 000, 100, 010, 110, 001, 101, 011, 111.

**Lemma 4.1** (local frozen-pair criterion). *Let  $s = (a, b, c)$  and  $t = (d, e, f)$  be two consecutive blocks. Their boundary is compatible with a frozen configuration precisely as follows:*

| $s$           | $\iota(s)$ | allowed next blocks $t$ |
|---------------|------------|-------------------------|
| 000, 100, 001 | 0, 1, 4    | all eight blocks        |
| 010           | 2          | blocks with $d = 0$     |
| 011, 111      | 6, 7       | blocks with $d = 1$     |
| 110, 101      | 3, 5       | no block                |

Equivalently, in the index order above the adjacency matrix is

$$(1) \quad M = \begin{pmatrix} 1 & 1 & 1 & 1 & 1 & 1 & 1 & 1 \\ 1 & 1 & 1 & 1 & 1 & 1 & 1 & 1 \\ 1 & 0 & 1 & 0 & 1 & 0 & 1 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 1 & 1 & 1 & 1 & 1 & 1 & 1 & 1 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 1 & 0 & 1 & 0 & 1 & 0 & 1 \\ 0 & 1 & 0 & 1 & 0 & 1 & 0 & 1 \end{pmatrix}.$$

*Proof.* After the aligned layer acts on  $s$ , write its output as  $(a, \beta, \gamma)$ . The gate definition gives

$$(\beta, \gamma) = \begin{cases} (b, c), & a = 0, \\ (c, b), & a = 1. \end{cases}$$

The shifted gate across the boundary has control  $\beta$  and targets  $\gamma, d$ . Its output on the three positions originally occupied by  $b, c, d$  is

$$\begin{cases} (\beta, \gamma, d), & \beta = 0, \\ (\beta, d, \gamma), & \beta = 1. \end{cases}$$

For these coordinates to be frozen, the displayed triple must equal  $(b, c, d)$ .

If  $a = 0$  and  $b = 0$ , then  $\beta = 0$  and  $\gamma = c$ , so every next block is allowed; these are 000 and 001. If  $a = 1$ , the equality  $\beta = b$  first forces  $b = c$ . The case  $b = c = 0$  gives 100 and again allows every next block. When  $\beta = 1$ , the target swap is frozen exactly when  $d = c = \gamma$ . For 010 this requires  $d = 0$ ; for 011 and 111 it requires  $d = 1$ . Finally, 110 and 101 fail  $\beta = b$  and admit no successor. The coordinates  $e, f$  of the next block do not enter this boundary gate. Translating the resulting conditions through  $\iota$  gives (1).  $\square$

**Theorem 4.2** (transfer description). *A ring configuration with block word  $(s_0, \dots, s_{m-1})$  is fixed by  $T_m$  if and only if*

$$M_{\iota(s_j), \iota(s_{j+1})} = 1 \quad \text{for every } j \in \mathbb{Z}_m.$$

Consequently,

$$(2) \quad f_m := |\mathcal{F}_m| = \text{tr}(M^m), \quad m \geq 1.$$

*Proof.* For a fixed  $j$ , lemma 4.1 is equivalent to restoration of the three coordinates  $b_j, c_j, a_{j+1}$  after the full two-layer update. As  $j$  runs through  $\mathbb{Z}_m$ , these triples partition all  $3m$  output coordinates. Thus every adjacent pair is allowed if and only if every coordinate is restored, proving the first assertion.

The diagonal entry  $(M^m)_{q,q}$  counts length- $m$  walks in the adjacency graph that start and end at block symbol  $q$ . Summing over  $q$  counts exactly the labelled cyclic block words satisfying every boundary constraint. This sum is  $\text{tr}(M^m)$ .  $\square$

## 5. CLOSED COUNT, ENTROPY, AND SPATIAL ZETA

The  $8 \times 8$  presentation is deliberately kept explicit because it makes the all-size count auditable. Its apparent dimension collapses spectrally.

**Lemma 5.1** (spectrum of the transfer matrix). *The transfer matrix in (1) has rank two and characteristic polynomial*

$$\det(\lambda I - M) = \lambda^6(\lambda^2 - 5\lambda + 3).$$

*Its nonzero eigenvalues are*

$$\lambda_{\pm} = \frac{5 \pm \sqrt{13}}{2}.$$

*Proof.* Let  $u = (1, 0, 1, 0, 1, 0, 1, 0)$  and  $v = (0, 1, 0, 1, 0, 1, 0, 1)$ . Every row of  $M$  is one of  $0, u, v, u+v$ , and both  $u$  and  $v$  occur, so  $\text{rank } M = 2$ . The eigenvalue 0 therefore has algebraic multiplicity at least six. Directly from (1),

$$\text{tr } M = 5, \quad \text{diag}(M^2) = (4, 5, 3, 0, 4, 0, 1, 2), \quad \text{tr}(M^2) = 19.$$

If  $\alpha, \beta$  denote the remaining two eigenvalues, counted with algebraic multiplicity, then

$$\alpha + \beta = 5, \quad 2\alpha\beta = (\alpha + \beta)^2 - (\alpha^2 + \beta^2) = 25 - 19 = 6.$$

Thus  $\alpha\beta = 3$ , so they are the two roots of  $\lambda^2 - 5\lambda + 3$ . Their product is nonzero, which also shows that the zero eigenvalue has algebraic multiplicity exactly six.  $\square$

**Theorem 5.2** (closed frozen-state count). *For every  $m \geq 1$ ,*

$$(3) \quad f_m = \lambda_+^m + \lambda_-^m = \left(\frac{5 + \sqrt{13}}{2}\right)^m + \left(\frac{5 - \sqrt{13}}{2}\right)^m.$$

*In particular,  $f_1 = 5$ ,  $f_2 = 19$ , and*

$$f_m = 5f_{m-1} - 3f_{m-2} \quad (m \geq 3).$$

*Proof.* By theorem 4.2 and lemma 5.1, the trace of  $M^m$  is the sum of the  $m$ th powers of its eigenvalues, counted with algebraic multiplicity. For  $m \geq 1$  the six zero eigenvalues contribute zero, yielding (3). Both nonzero eigenvalues satisfy  $\lambda^2 = 5\lambda - 3$ ; multiplying by  $\lambda^{m-2}$  and summing the two identities gives the recurrence. Substitution gives the stated initial values.  $\square$

Define the two-sided frozen subshift over three-bit block symbols by

$$X_{\text{fr}} = \{(s_j)_{j \in \mathbb{Z}} : M_{\iota(s_j), \iota(s_{j+1})} = 1 \text{ for every } j\},$$

and let  $\sigma$  be the left shift by one block. Standard finite-type shift facts identify  $|\text{Fix}(\sigma^m)|$  with  $\text{tr}(M^m)$  [3].

**Corollary 5.3** (frozen entropy and spatial zeta). *The topological entropy of the block shift on  $X_{\text{fr}}$  is*

$$h_{\text{top}}(\sigma|X_{\text{fr}}) = \log \lambda_+.$$

*Normalized per original binary site, the entropy is  $\frac{1}{3} \log \lambda_+$ . The Artin–Mazur zeta function of this spatial block shift is*

$$\begin{aligned} \zeta_{\text{fr}}(z) &:= \exp \left( \sum_{m \geq 1} \frac{|\text{Fix}(\sigma^m)|}{m} z^m \right) \\ &= \det(I - zM)^{-1} = \frac{1}{1 - 5z + 3z^2}. \end{aligned}$$

*The identity holds as a formal power series and analytically for  $|z| < \lambda_+^{-1}$ .*

*Proof.* The entropy of a finite-type shift presented by  $M$  is the logarithm of the spectral radius of  $M$  [3]; lemma 5.1 gives  $\rho(M) = \lambda_+$ . One block contains three original sites, giving the per-site normalization.

For the zeta identity, use the formal matrix expansion

$$-\log \det(I - zM) = \sum_{m \geq 1} \frac{\text{tr}(M^m)}{m} z^m.$$

Then apply theorem 4.2 and  $\det(I - zM) = (1 - z\lambda_+)(1 - z\lambda_-) = 1 - 5z + 3z^2$ . The same expansion converges analytically inside the reciprocal spectral radius.  $\square$

*Remark 5.4* (the zeta boundary). The variable  $m$  in  $\zeta_{\text{fr}}$  is a spatial block period. For a fixed ring size, the temporal Artin–Mazur zeta function of  $T_m$  would instead require every count  $|\text{Fix}(T_m^k)|$  over time iterates  $k$ . None of those temporal zeta functions is asserted or inferred here.

## 6. EXACT FINITE CONTROLS AND TEMPORAL OBSERVATIONS

The accompanying dependency-free script implements the literal gate rather than the transfer characterization. For each  $m = 1, \dots, 6$ , it exhausts all  $2^{3m}$  states, checks both layer involutions, the explicit inverse and reversing symmetry, Hamming-weight conservation, bijectivity, and the equivalence between literal fixed points and the SFT constraint. It also compares the fixed count with (2) and (3).

| $m$ | states  | $f_m$ | maximum temporal period | observed period set                                  |
|-----|---------|-------|-------------------------|------------------------------------------------------|
| 1   | 8       | 5     | 3                       | 1, 3                                                 |
| 2   | 64      | 19    | 4                       | 1, 2, 3, 4                                           |
| 3   | 512     | 80    | 8                       | 1, 2, 3, 4, 6, 8                                     |
| 4   | 4,096   | 343   | 18                      | 1, 2, 3, 4, 6, 8, 10, 12, 14, 16, 18                 |
| 5   | 32,768  | 1,475 | 32                      | 1, 2, 3, 4, 6, 8, 10, 12, 16, 18, 30, 32             |
| 6   | 262,144 | 6,346 | 74                      | 1, 2, 3, 4, 6, 8, 10, 12, 16, 18, 24, 30, 32, 54, 74 |

These are exact finite computations, not samples. Their total state count is

$$\sum_{m=1}^6 2^{3m} = 299,592.$$

The increasing maximum periods motivate further questions about the temporal permutations, but the table alone neither proves unbounded periods nor identifies a temporal zeta function.

## 7. DISCUSSION

The frozen set of this staggered conservative permutation is simpler than its finite temporal cycle structure. The local fixed equations reduce to a rank-two transfer matrix even though the natural block alphabet has eight symbols. That reduction supplies an all-size theorem: exact counts, a second-order recurrence, entropy, and a rational spatial zeta function.

Several temporal problems remain outside the present contract. The data do not prove an unbounded-period family, a closed cycle index, mixing properties of temporal trajectories, or integrability. Any future temporal claim must be proved independently and compared directly with the Fredkin staircase and other constrained-swap automata. The present conclusion is narrower: for the map in [definition 3.2](#), the frozen configurations have the exact finite-type presentation and spatial enumerative invariants proved above.

#### REPRODUCIBILITY

The file `code/verify_fredkin.py` uses only the Python standard library. It prints the full finite cycle types and an instrumented count of executed checks. The computation is a regression control; the all- $m$  statements rest on the proofs in [lemmas 4.1](#) and [5.1](#) and [theorem 4.2](#).

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